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Composition and Perception in Spatial Audio

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Composition and Perception in Spatial Audio

Abstract: This article discusses the advantages of spatial audio, in general, followed by strategies for applying spatial components to composition. The discussion then looks ahead to questions that may be solved by future implementations of spatial software and hardware. Despite the fact that technical systems for spatial audio have been in use since the 1950s, spatial concepts have not been widely integrated into the compositional process. This is because they involve a complex interaction of several phenomena, all of which play a role in the construction and perception of music. This article presents an analysis of the advantages of spatial audio for perception and provides examples of the decomposition of sonic material with the help of spatial properties, as well as a discussion of limitations in spatial construction and perception.

Archival strategies for spatial audio are also briefly discussed.

Shortly after being appointed director of the Institute for Music und Acoustics (IMA) at the Zentrum für Kunst und Medientechnologie (ZKM) in 2003, I began to work on the Klangdom project together with the team of the IMA. This involved developing and selecting the spatialization technology best suited to the project, as well as conceptualizing and organizing the software development and the installation of a high-density loudspeaker array (HDLA; see Brümmer et al. 2014 for more background on the Klangdom). As one of the earliest continuously accessible HDLA environments (see Figures 1 and 2), it initiated numerous experiences and practical examples of composition and perception that need to be discussed in this special issue of *Computer Music Journal* focusing on HDLA systems.

The Klangdom was operational as early as 2004, but it had not yet reached its final stage of expansion. This was a time in which artistic projects were begun by numerous guest artists at the ZKM, as well as by the author. The knowledge gained from observations and considerations of the practical work carried out form the basis of the facts outlined in this article.

The diffusion methods discussed here relate to vector base amplitude panning (VBAP, see Pulkki 2002) and distance-based amplitude panning (DBAP, cf. Lossius, Baltazar, and de la Hogue 2009), along with Wave-Field Synthesis (WFS, Berkhout 1988) and Ambisonics (Gerzon 1974). In general, the geometrical forms of HDLAs may be described as domes or rows. Systems of greater complexity contain several cupolas or domes, such as "The Morning Line," the sound pavilion produced by the Thyssen-Bornemisza Arts Contemporary Foundation and housed at the ZKM (http://zkm.de/event/2013/09/the-morning-line), the 4DSound system (http://www.4Dsound.net), and the Acousmonium (Prager 2012).

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Advantages of Spatial Audio

There are a number of advantages in using an HDLA, as opposed to only a few speakers, to listen to sound. Some of these advantages depend on the kind of projection technology, whereas others apply to almost every sound system. The following description of the most important psychoacoustic and acoustic phenomena outlines the advantages of spatialized sound perception. Some of the phenomena contain great potential for an increase in differentiation in the perception of complex sound scenarios. Overall, it is clear that the simultaneous occurrence of all of these phenomena has brought about a dramatic change to the way sounds are heard. To analyze these phenomena, it is necessary to discuss each of the specific effects individually.

To begin with masking, this effect occurs when quiet sounds are drowned out by loud sounds. When signals are summed together during mixing, the masking effects produced are determined and perceived by listeners in the same way, regardless of their position in the room. In spatialized sound environments, the summing of signals happens

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Figure 1. The Minidome spatial studio at the Institute for Music and Acoustics (IMA) at the Zentrum für Kunst und Medien (ZKM), Karlsruhe. (Photograph by Bernhard Sturm, copyright 2006 ZKM/Karlsruhe.)

because of the constellation of sound sources with respect to the listening position. If the listening position is not changed, the masked and masking signals interact in a variety of ways.

For example, if a quiet tone sounds spatially separated on one side of the listener and a loud tone sounds on the other, the listener will be able to hear both sounds despite the difference in volume. If these sounds were to be mixed together, the masked sound would be inaudible.

Masking also happens within spatial sound systems, but far less often and in ways that are more variable. For instance, moving the head can reveal masking (Neukom 2005, p. 627), and artists such as Maryanne Amacher have systematically used a phenomenon known as *otoacoustic emission*. Jonathon Kirk (2010) describes otoacoustic emissions as "sounds that arise in the inner ear and which are related to the process of amplification in the cochlea." The use of certain frequency relations triggers difference tones. If the source signals are emitted from different locations, moving the head slightly can modulate the difference tones.

It is also possible to separate the information under certain circumstances, because the masking process involves the combination of several pieces of acoustic information. By blocking out all other sounds present in an auditory scene, it is possible to concentrate on just one of them. The ability to focus this way is commonly referred to as the "cocktail party effect" (Cherry 1953). An important contributing factor for focusing is, however, the distribution of sound sources throughout the space (Mesgarani and Chang 2012).

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Figure 2. The Klangdom in the IMA Kubus. (Photograph by Bernhard Sturm, copyright 2006 ZKM/Karlsruhe.)

The separation of certain pieces of audio information requires not only spatial separation of the sound sources. The process is aided if the structure of the sound has as many individual qualities as possible—one of which is a fluctuating phase, the result of even a tiny movement of a sound in space.

Even a continuous movement of sounds makes them seem more realistic. This effect is achieved through the movement of an emerging fluctuation or shift in phase. This fluctuation occurs in most sounds heard in nature. Through slight shifts in all sounds, a natural sound environment can be reproduced.

In a natural auditory scene several individually moving sounds would be present at the same time.

It is possible in every spatial environment to achieve a gradation of the depth of sound events in terms of distance. As a result, a sense of perspective for each individual listening position is created. This can be influenced by changing the listening position of the listener, producing the possibility of influencing sound events through distinct movement. The sound event can then be designed interactively, as listeners are able to move closer to certain sounds and distance themselves from others. Indeed, when projecting sound in an HDLA, it is no longer possible to achieve an ideal mix of the sounds at their optimum level, as is usually the case with stereo projections. Sounds can be heard in considerably greater detail, however, by focusing,

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turning the head, or even moving physically closer toward certain sound objects. A composition can be perceived in a different way from seat to seat or by walking through the HDLA environment. This results in the activation of the listening process. The listener is now able to interact with the auditory scene. In addition, compositions can sound different, depending on the listener's seat or position, and a second hearing of a work can lend it further intensity.

The individual quality of perception at each position in the concert situation is enhanced when reverberation is implemented through acoustic ray tracing (Schroeder 1969; see also Blauert 1997). Through the dispersal of sound within a room, reflections of the sound are created that combine the phenomena of echo and reverberation. With traditional methods, the reflections cannot be rendered in a direction-dependent manner. With stereo, two rendering channels are sufficient. With WFS systems and sound-dome rendering, reflections can be triggered from different directions, in a manner analogous to ray tracing. In this way, it is possible to render not only the acoustics but also the structure of the room in detail, and the listener receives a more accurate picture of the room characteristics. A simplified version of the acoustic modeling of the reflections can be produced with the Spat library (Carpentier, Noisternig, and Warusfel 2014). It is conceivable that this method could be expanded to check the degree of detail to which the process could best be reproduced to improve spatial perception.

Regarding the enhancement of the perception through spatial environments, it is true that the usable space itself is enlarged up to the entire area covered by the speaker array.

All reproduction systems for surround sound expand the area available for positioning sound. Systems using VBAP and DBAP make the space under the cupola available for sound dome systems, and WFS systems do the same for the space behind the speaker rows. In both cases, a larger staging space is created, in which more sounds may be positioned for clear identification. It is worth noting here that the placement of sound objects above the listeners works particularly well: No acoustic shadows are produced by nearby objects or people, and the location capabilities in these areas are excellent (since they can occur binaurally).

Along with the enlargement of space comes a new quality of perception. In a standard stereo setting, the user hears sound objects from a position outside the observed region. When listening to sound projected from an HDLA, the listener becomes part of the surrounding spatial environment and observes it from a position in its midst. This has a significant impact on the intensity of participation.

The playback of sound within the room produces an immersive environment, and this changes the effect on the listener. The immersion increases the feeling of intensity and creates sounds that correspond more fully to those that normally surround us. The soundscape heard this way can no longer be described as an object, but as an environment. Immersive sound perception is clearly more impressive than a simple, frontal audition. Applying immersive sound perception to acoustic instruments performing in a concert can be realized by placing the instruments around or among the audience.

Applications of Spatial Composition

Regarding the application of strategies for the spatial simulation of sound, it is necessary to describe some of the possibilities as well as their specific advantages and disadvantages. Many of these strategies have already been applied and extensively tested, but composers' intentions must be clearly identified before starting the compositional process. Combinations of these strategies are possible, and new strategies may also develop. The strategies mentioned here should provide a guide and suggest a few options and solutions.

Deconstruction of Acoustic Information through Space

The spatial positioning of sounds can interfere with the amplitude of sonic information and can impose its own structure on existing ones. In instrumental

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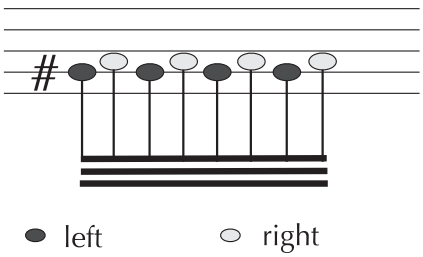
Figure 3. Trill separated in space, available as Audio Example 1. (URLs for the audio samples are given in the article text.)



music, structuring is usually done through the articulation of the tones (e.g., staccato, legato, or accents) or through the instrumentation, specifically through the structuring of different timbres. A structural change to the acoustic information can be achieved through the spatial positioning of tones, and one can even produce an ambiguous perception, as is explained in the following example (taken from Clarke 1999).

A trill is first rendered monophonically, followed by the playing of a lower tone on the left speaker and the higher tone on the right speaker (see Figure 3 and Audio Example 1 at http://www.mitpressjournals.org/doi/suppl/COMJ_a_00402-Brummer). The trill is now easily audible at the center. If the listener stands to the right or left of the center, the trill will become a repeated tone with a slight delay and a quieter second note. The existence of three different listening variations of the same signal is evidence of the deconstructive effect achieved through the positioning of signals in space.

The next step is to play the notes of the trill over the right or left speaker in an irregular rhythm, with the result that a completely new pattern is created. Here is a concrete example: If G#–A is played on the left speaker, G# on the right, A–G# on the left and A again on the right, followed by G# on the left and A on the right (see Figure 4), a kind of metarhythm is introduced via the spatial distribution. This counters the regular pattern of the trill and competes with its structure.

This creates an interesting relationship between the two structures, which is, however, strongly dependent on the position of the listener. If the listener stands at the exact midpoint between the two sources, the regularity of the trill becomes more

Figure 4. Trill separated in space applying rhythmic properties, available as Audio Example 2.

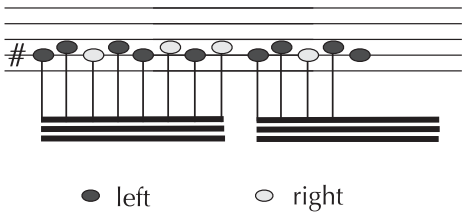


Figure 4

Figure 5. Pattern heard from the left side.

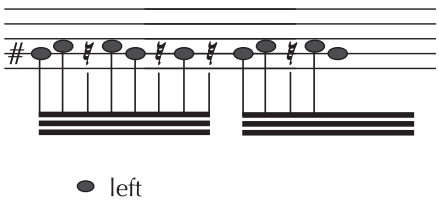


Figure 5

Figure 6. Pattern heard from the right side.

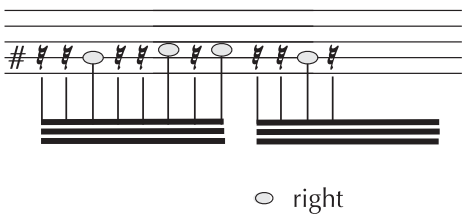


Figure 6

apparent to the listener, while on the left side one hears the following rhythm (notes written in capital letters are louder, and those written in lowercase letters are quieter):

G# A g# A G# a G# a, G# A g# A G# a G# a

If the notes playing from the right speaker are left out, the listener will hear the rhythmic pattern shown in Figure 5.

Whereas one would hear the following structure on the right side:

g# a G# a g# A g# A, g# a G# a g# A g# A . . .

By omitting the notes from the other speaker, the structure displayed in Figure 6 can be heard.

In this way, three drastically different versions of the same material are presented, and how they are heard depends on the position of the listener.

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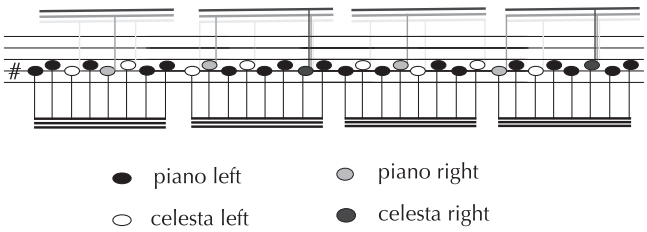
Figure 7. Trill performing a four-level fractal structure, available as Audio Example 3.



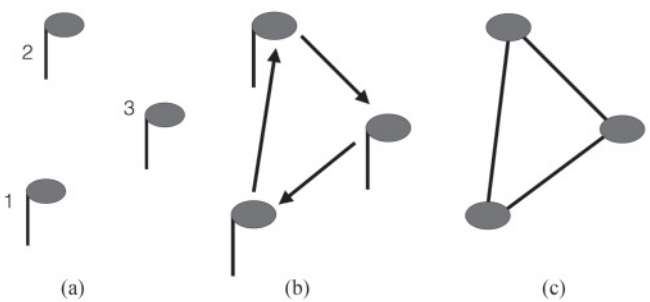
It is possible to extend this technique to create a "fractal" structure with four levels using only two instruments and two speakers. This trill is displayed in Figure 7 (and Audio Example X at http://www.mitpressjournals.org/doi/suppl/COMJ_a_00402-Brummer).

Additionally, a placement of sound in space provides the opportunity to produce geometric shapes. If three short notes are played relatively quickly one after another, they are perceived as a group, condensed into one phrase and felt as though they correspond to each other. If these notes are positioned in a triangle in the space, the audience will perceive this geometric shape, articulated by sound.

Impact of Speed on Spatial Perception

In all of the examples set out here, the way the sounds are perceived is affected by the speed in which they are presented. The basis of this fact is described in considerable detail in the results of the research of auditory scene analysis (Bregman 1990, ch. 2, section "Temporal Relations," see also "Spatial Location"). As such, three speed-dependent modes are conceivable. When the sounds are played very slowly one after another, three discrete notes are detectable at three points (see Figure 8a). When they are played more quickly, one can tell that the notes form a self-contained group that is also distinctly recognizable as a triangle. As the tempo increases, the sounds appear to move towards their respective following location, although technically they are still static (see Figure 8b). When the speed is increased significantly, the listener experiences these three sounds as one static sound with a certain spatial expansion (see Figure 8c).

Figure 8. Speeding up three sounds: discrete (a), moving (b), and static (c). Available as Audio Example 4.



The sequence of this perception process can be modeled on the following test setup. In this instance, however, it is necessary to use a trill with the lower and higher notes positioned to the left and right, respectively. When the trill is sped up, the listener can detect all of the aforementioned stages either discretely, grouped together, grouped together with movement, or as static areas.

In these examples, the influence of space and of the positioning of sound events in space on the perception of the structure is verified. This phenomenon can be used intentionally for composition, as the example with the trill demonstrates. All of the phenomena mentioned here also occur without any planning, and they arrange the material according to the aforementioned conditions including structure, speed, place, and listening position. They also indicate the perceptual potentials that lie in varying the listener's position, as well as showing the diversity of materials produced as a result of that variation in listening position. It has also been established that the spatialization of sounds can accomplish tasks similar to phrasing and instrumentation, for example, and spatialization can be used in a multitude of additional ways. It goes beyond pure structuring of information and makes interactive listening possible.

Algorithmic Distribution

The use of algorithmically controlled spatialization mostly involves coordinating a variety of sound objects. This method has proved itself particularly effective in granular synthesis, where the room holds a large variety of grains. These can be mapped as

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clouds or swarms and set in motion until the grains settle in a random configuration. This is where algorithms come into play, capable of organizing the groups into a hierarchy so that they appear uniformly in the desired order. Rama Gottfried impressively demonstrated this in his 2012 work *Fluoresce*, for cello, higher-order Ambisonics, and WFS system, performed in the Klangdom during the inSonic2015 conference, in which he placed the grain clouds in real time in different layers in the available space. In the program notes for this piece, Rama Gottfried (2012) writes: "Just as complex timbre, when understood spatially, can be described in terms of focal plane—fore, mid, and background layers—complex spatial organization as gestalt can be understood as timbre." This principle of organizing many individual pieces enables the creation of perceptual unity, which in turn allows for a better arrangement of complex sound elements. Another example of an algorithmically generated spatialization is a work created by Bruno Friedmann, where spatial positions are derived from the properties of sound generated by the audio descriptor function `ircamdescriptors~` in Max (Brümmer et al. 2014).

Agent systems are also noteworthy as a possible way of producing a decentralized pattern of order and have been used in several works. These systems are made up of sound objects whose movement and behavior are influenced by the position of the other objects. An example of such a rule might be collision avoidance in the context of a random pattern of movement.

Optimization of Simultaneous Structures

Although certain sound structures are developed using individual procedures, speeds, densities, and timbres, they need to be combined if they are to create a rich and interesting composition. In this process of combination, the limits of perception have to be accepted, though they can be widened through the integration of spatial properties.

Matching particular materials with particular positions is another possible way of using spatial information to contribute to a composition. This need not necessarily involve didactic aids in the perceptual process, which can provide such a connective framework. Links between materials can be revealed, be they related to the local space or to a more distant environment. Structural combinations may also be of help here. If, for example, four fast and four slow structures are played simultaneously, they could be arranged so that each of the fast structures is positioned in four corners, with the four slow structures placed at four central locations, shifted 90 degrees from the materials located in the four corners. This helps to achieve a better level of perceptibility of the different sound materials.

Karlheinz Stockhausen's *Oktophonie*, composed in 1991, demonstrates the assignment of the content of sound qualities to specific locations in the space. Eight loudspeakers, placed in the vertices of the room, describe six themed rectangular spaces. Stockhausen assigned certain sounds or material to the front, back, left, right, roof, and ground spaces, each defined by four speakers. Although the sounds appear only in the assigned squares, he introduces a correlation between sounds and their position and movement. In this way, the arrangement of the space serves as a metaphor for the context and the interconnected nature of the different sounds.

Mapping

The use of tonal positioning as a structural method opens up new possibilities for the parametric use of the space. The room's conceptual use becomes endowed with an additional parameter level that is applicable for the composition process.

The mapping of other parameters onto the spatial position is a highly efficient and variable way of exploiting the space in a complex, yet easily understandable way. Specific pieces of information, such as details concerning pitch and position, are duplicated and can therefore be perceived more efficiently. An additional function can be applied in the mapping between a parameter and its spatial appearance. An exponential function may distort the pitch information, avoiding the doubling of the parameter. A complex algorithm may introduce a more interesting relation in the mapping between

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pitch and position (Brümmer et al. 2014). Even when translating real movements onto the acoustic space, it is necessary to map the controller data space onto the sound space.

Motion as Enhancing Realistic Acoustic Presentation

In experiments at the ZKM, we discovered relatively quickly that acoustic motion had a positive impact on the realistic perception of sounds. The quality of this motion played less of a role than the difference between stasis and motion.

By way of explanation, consider that there are hardly any truly static sources of sound. Our sense of hearing mainly notices sounds that are in some sort of motion, so its sensitivity is geared towards dynamic modulation of phases. From these considerations, we can deduce that a sound that is in some kind of motion reflects this quality better than a static sound. This motion should consider not only the phasal movements contained within the signal but also the phasal mixtures generated by the reflections of surfaces. Proof of this phenomenon can easily be found by taking a phase-modulated sinusoid as a starting point. Another way to demonstrate this would be to record a moving sinusoid, revealing the mixture of reflections and phase modulations produced.

Statistical Distribution of Sounds in Continual Motion

If multiple tracks are used, each containing densely layered textures, it makes sense to ensure that these tracks are well distributed throughout the space and do not overly interfere with one another. To this end, care should be taken to avoid overloading individual loudspeakers. Should it also be necessary to keep these distributed sounds in continual motion to achieve a particular feeling of naturalness, a strategy needs to be developed to avoid multiple small objects gathering in one area. The key here is to use rotations or spiral movements of varying speeds. It would also be possible for each track to be linked to more individual movements, governed by intelligent decisions about constellations, as already present in the aforementioned agent systems.

Linear and Nonlinear Motion

The speed of the movements must be varied if they are to portray expressivity or a gestural quality. Markedly exponential increases or decreases in speed can lend the motion a gestural or natural character. John Chowning's quadraphonic work *Turenas* is a particularly good example of this. Variable-speed qualities can be achieved with editing software, which can control these in the form of segments, allowing chunks of movements to be sped up or slowed down (for an example, see the software Zirkonium MK3).

Gestural Interpretation

New projection techniques allow sounds not only to be positioned but also to move about easily in a two- or three-dimensional space. This option was put into practice as early as 1958, in the Philips Pavilion at the Brussels World Fair, but was refined in 1970 in a spherical auditorium in Osaka, with the aid of a *Rotationsmühle* (literally, a "sound mill"; cf. Hein 2004). Here, loudspeakers could be patched onto a path of motion and then continually controlled with the help of a rotary connector. At the time of the spherical auditorium, it was only possible to transfer the movements onto one channel using the *Rotationsmühle*, whereas modern technology today allows for transfer to 80–90 channels at a time. In this regard, it is important to take account of the differences between the above-mentioned linear and nonlinear motion.

Linear motion is marked by a certain flexibility, although it may lack gestural intensity. It can be extremely useful in the static distribution of sounds in continual motion, because it tends not to draw any particular attention to itself. Nevertheless, the introduction of planned movements can lend an expressive gestural quality with increases or decreases in speed. Just as expressivity can be

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created by volume and rhythmic speed, motion speed can also generate a significant expressive impression.

Input controllers can also be used alongside the mapping or construction of movements. Spat, Zirkonium, and other spatialization software generally use Open Sound Control protocols to transfer information about the controller to the data processor and the motion software.

Intuitive Expression of Movement

Motion speeds can be determined by various input controllers, for example, a mouse or a joystick. This permits positions and speeds to be modeled and saved. This method is advantageous when it comes to setting sounds in motion as an interactive reaction to what has been heard. Further possibilities can be created when the data are then edited in large quantities. Generally speaking, paths of movement are marked out by a series of dots, which can only be partially modified, however. It is possible to group these dots together before scaling or rotating them. More possibilities emerge if a movement is defined by Bézier curves. In this case, the shape of the motion can be altered, rescaled, and rotated, post facto, with the aid of "handles." When using plotted data, there is a danger of having an overly subjective application of movement and sacrificing movement as one parameter on a par with other parameters. This can be prevented by using motion modules (see Zirkonium MK3). In addition, there are methods by which intuitive and conceptual strategies can be simultaneously utilized. Combining amplitude indicators with spatial trajectories gives a good indication where a certain sound gesture appears in space, as shown in Figure 9.

In his work *Mittwochs-Gruss* (2003), Stockhausen used interactive control technology in the form of a touchscreen, allowing him to move and store sounds during the performance. Stockhausen achieved the motion choreography through a mixture of data controlled fully and subjectively with the help of a 2-D controller combined with a motion concept, similar to what he had used for *Licht-Bilder*. Modules, such as a circle or line, are notated along with direction, angle, and time. This score is then transformed in real time as the player of an instrument plays the relevant musical notes after rehearsing them. When Stockhausen listened to *Mittwochs-Gruss* for the first time on concert speakers in the Kubus at the ZKM, he informed me about this semi-interpretive process of transposing his polyphonic notation.

Further possibilities for the development of gestural qualities are to be found in the transfer of real movements to the sound object. Wireless- or camera-based motion sensors can be used for this purpose. The recorded data have the advantage of a gesturally consistent expressivity but the disadvantage of subjective materiality, which is formally difficult to structure. They are firmly established post-recording and can be heavily edited. For this reason this process is mostly used in real time, focused on the attraction of transferring bodily movement, in particular dance, to the sonic level.

Immersive Depiction of Sound Objects

Recorded sound objects have the potential to possess a variety of vibrational states in various parts of the instrument. This is why electric guitars mostly have two or more pickups, each with its own distinctive timbre and phasing. This principle can be transferred to other instruments and then diffused through a surround-sound system. Imagine a gong or a piano recorded with 20 pickup microphones in different positions. What is particularly interesting is creating a physical model whose vibrations are recorded in various positions, similar to what Claude Cadoz has done in a number of his works. When these signals are echoed throughout space, they produce a special kind of experience that makes listeners feel as though they are standing in the middle of the instrument. The gong, the piano, or the physical model represents an immersive vibration environment, as can be heard, for example, in the immersive compositions *Helios*, by Claude Cadoz, or in *Spheres of Resonance*, by the author.

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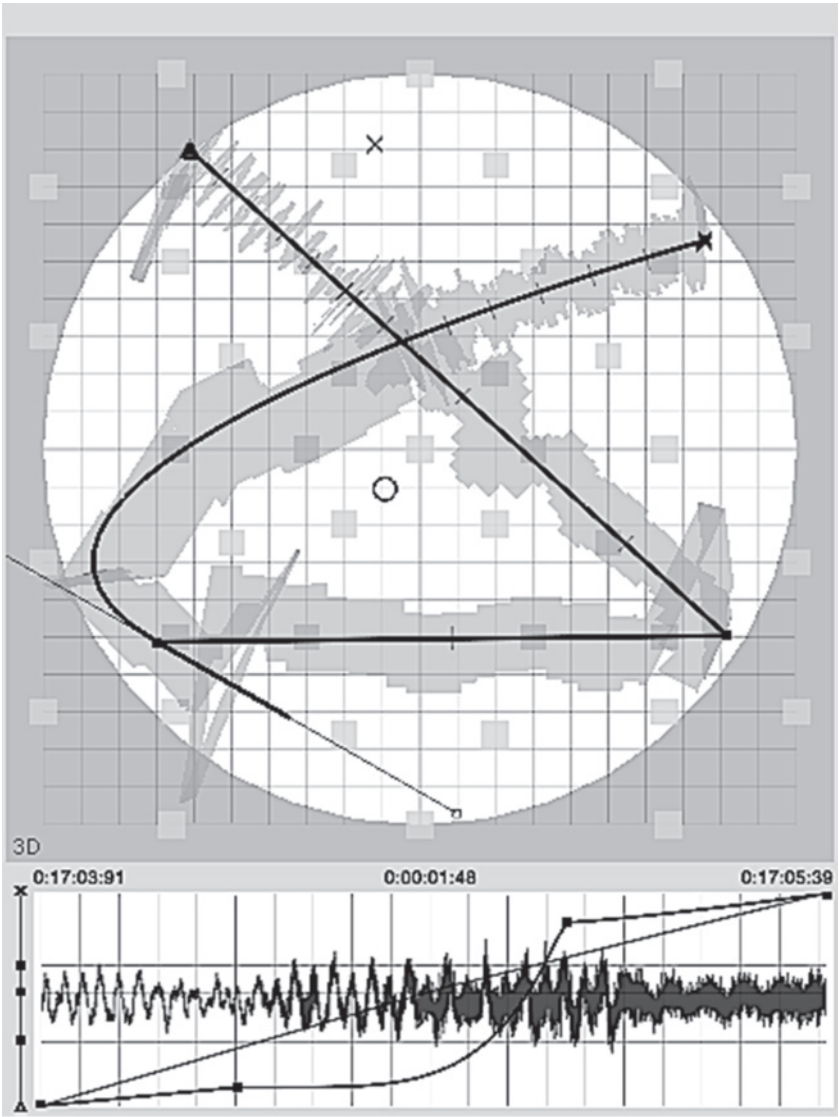
Figure 9. Screen shot of Zirkonium MK3's Motion and Speed Editor, which uses Bézier curves in conjunction with the amplitude display of a given sound segment. Zirkonium MK3 was designed by Chikashi Miyama.



Problematic Aspects

Several problems occur applying movements or placements of sounds in HDLAs. The desired spatial impression defined by parameters can differ from the perceived impression, sometimes in significant ways. This may have psychoacoustic, physical, or other reasons. Part of the practical experience and learning process is to develop a sense of when not to expect impossible results from a given sound and acoustic situation.

The Contradiction between Applied Parameter and Acoustic Result

A problem relating to spatial scores can arise in a possible contradiction between editing and

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perception. Most editors shift sound files and not sound events. The movement occurs independently from the sonic substance of the sound file. If this contains a continuous sound, a congruence is created between the control data and the sound impression. If it contains a temporally discrete rhythmic events of short duration, however, such as a repetition or more rhythmically complex events, the continuous movement of the sound file results in a distribution of the events to apparently fixed positions in space, without making the initiating movement comprehensible. Through rapid repetitions in combination with a slow-motion path, the movement may still be recognizable. This impression stands out from even more significant events that take place with greater temporal distance and appear to sound discrete and to be placed in a particular position. An increase in the speed of the movement in the space editor in this case would only be heard as a different spatial distribution of individual events. In this situation, the parameters of the movement do not produce an adequate description of the result heard. In response to this, a function was introduced in Zirkonium MK3 that scans for pauses so that a separate parameterization of position or motion can take place for every region above a certain amplitude threshold. This inconsistency between parameters and sound result cannot be entirely resolved.

Limits to the Perceptibility of Shifting Sounds

A serious problem with the gestural application of motion lies in the quantity of observable objects. In a way similar to visual media, the listener can hear only one object in detail and two or three objects with adequate acuity. In the perception of moving voices, a comparable limit exists. A three-voice fugue provides the maximum level of detail for perceivable voices. A four-voice fugue indicates in its very composition that certain voices are redundant, and five or six voices can hardly be perceived in any detail (McDermott and Oxenham 2008, pp. 457–458). In this case, all but the most significant characteristics are filtered out of the event, while the rest of the information is more or less ignored. In many situations, various voices merge with one another to form a single voice. Polyphonic listening is, however, a skill that can be improved, to a certain extent, through training. This is possible particularly when the amount of perceivable information in sound and pitch is reduced, as is the case in chromatic scales with timbres of familiar instruments.

Nonlocatable Sounds

One dilemma posed by spatialization is that many composers have unrealistic expectations of such a system—namely, that it should result in each and every sound being clearly located—and criticize the system for not being capable of achieving this. Consequently, they are often unhappy with the system's results. Closer inspection of the reasons for the failure to spatially perceive individual sounds reveals, however, that the root of the problem is actually the nature of the sounds themselves, as well as the fact that the listener may not be aware of all that is to be heard. Sounds without high-frequency components, or continuous tones, can be difficult to locate. In addition, low-frequency components of a complex sound can easily drown out the high-frequency elements.

But what happens when the listener does not clearly detect the location of sounds within the space? The fact that a sound may have qualities that make it difficult to locate is often not obvious to the listener but, instead, results in the sound being all the more difficult to hear. The fact that the sound's location or motion cannot be resolved goes unnoticed. The location of a sound that is difficult to perceive can only be determined by focusing one's attention on that particular sound. This phenomenon must be made clear to composers working with surround-sound systems, so that they can adjust their expectations accordingly. Only then can the decision be taken as to whether to leave certain sounds as too difficult to locate or to optimize the sound material to prioritize location.

Archival and Cross-Platform Compatibility

A fundamental problem in producing surround sound is cross-platform compatibility between

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the various software platforms and simulation techniques used. In addition, questions about archiving remain open.

One solution to this problem is to apply movements to an existing spatial environment and to encode them into the sound files. This strategy has the advantage that only the speaker environments would need to be reproduced. This would, however, limit the reproduction of the work to a certain spatial environment and would hence prohibit the interpretation of the work using future developments in spatial reproduction.

An alternative solution is possible, if composers were to use algorithms to perform the movements in their works. These algorithms can be stored, they can even be written down on paper. A disadvantage of this strategy is that a reimplementation of the program code would be necessary, should the programming language be no longer available. An algorithm is able to store and recompute complex movements, but questions remain as to whether the reimplementation of the algorithm will cause problems in the future and whether an algorithmic generation of movement causes a limitation in a perceptually oriented process of composition.

When it comes to cross-platform compatibility, a protocol is needed to abstract the movements present and to restructure them into simpler units that are syntactically independent, and yet which contain content that is more useful. One extreme possibility would be to save the spatial positions in relatively short units of time. As in audio files, it would be possible to describe the most varied of movements and positions, as well as the number of simultaneous positions, without having to resort to modes of movement that are too syntactically rigid. At the same time, data formatted in this way could be used to archive surround-sound works, regardless of the rendering software and the spatialization process used. If each renderer could process this format, it would be possible in the future to reproduce and, where necessary, to standardize movement scores across different editors, Ambisonics, panning procedures, and WFS systems.

One disadvantage at this point is the fragmentation of the data, carried out in the interests of compatibility and long-term archival. If the score is subject to new interpretation, for example, recorded on paper or even subjected to analysis, then the parameters set by the composer, as well as the structural descriptors given by the editor, are irretrievably lost. The only solution therefore seems to be the archival, in parallel, of fragmented data and structurally coded data, provided by the editor software.

The SpatDIF project seeks to solve both problems. A multilayered solution offers a way of saving different formats in a variety of ways. On the one hand, certain simple units of movement can be saved, such as Bézier curves and, more recently, circles and spirals (Miyama, Schacher, and Peters 2013). On the other hand, it is also possible to use series of dots to portray movements with greater complexity, for example, those produced by sketches. It is now necessary to agree upon an exchange syntax and to put this into practice as swiftly as possible.

Vision

In a discussion with Stockhausen during the production of *Licht-Bilder* in Karlsruhe in October 2004, the composer shared with me his vision of a surround-sound system. He said he dreamed of a sound speeding through a room, directly past one's ears. His conception of sound was one that has, to this day, remained a dream. Even for modern systems, controlled by VBAP, DBAP, WFS, or Ambisonics, it is impossible to bring sounds so close to the listener's ear, although the "timereversal mirror" process with a WFS array has been able to achieve a remarkable impression of sounds moving nearer to the listener (Höldrich and Sontacchi 2005). The 4DSound system (http://www.4dsound.net) goes one step further, with loudspeakers equally spread out in the interior allowing sounds to appear at a distance of 2 m or less from a listener.

A far more realistic idea is the development of individual renderers for head-related transfer functions (HRTFs). This may currently require considerable effort, but will soon become much easier to implement. It requires a new production

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channel, since the rendering process can only be carried out after obtaining the individual filters and the sound sources need to be available discretely until then. This is certainly the case with consumer formats like Dolby Atmos and DTS X, whereby the data for the HRTF can be loaded to the stereo system and the signal can be calculated in real time.

Microphones for recording spatial environments need to become easier to handle in the future before they can find wider distribution. Interestingly, there are already camera arrays for the recording of panoramic films available for the consumer market Why the wait for affordable hand-held microphone arrays?

The trend toward spatial listening is currently showing strong signs of further development. The acoustic spaces in cinemas and concert halls are equipped with improved speaker systems and the technical expertise to use them to better effect. There should be a greater effort to integrate spatial sounds and their compositional application into composing curricula. Additionally, spatial sound will continue to gain a foothold in the consumer market with the help of standardized playback systems.

Strategies for spatialization in real time are still a great area of experimentation. Here composers could work with systems that could implement specific rules as software agents for the distribution and movement of sounds, and so leave to performers the interpretive and gestural control over movement that occupies the foreground.

Using swarm algorithms, physical models, or other algorithmic organizational forms of large quantities of sounds, it is possible to play compositions live through 40 or more channels. This is happening now, though experience and routine are still lacking. There is room here for the development of experimental and practical strategies.

Developments in software should integrate aspects of the presentation of acoustic qualities in space into the software's interface. As well as this, new methods for defining trajectories or gestural movement must be developed, as has been happening with Zirkonium MK3 or Spat.

Composers should develop several versions of a composition for different surround-sound systems in the production process. The motion data could be exchanged between these systems with the help of a protocol compatible with SpatDIF and then adapted.

A protocol for the archival of motion information is essential to make possible the long-term, cross-platform storage of compositions.

Conclusion

Surround-sound pieces performed on HDLAs allow for a more interactive and flexible reception, resulting in a listening experience that is more active. In particular, virtual acoustic spaces can help complex sound settings become more interesting, transparent, and multilayered. Therefore, the listener is able to distinguish between many nuances of acoustic input, allowing a composition to unfold its full potential. Although there remains much room for growth within the practice of composition, it is worth noting that even now, one can find a wide range of works and ideas, and that classics of the electroacoustic repertoire can be adapted for such new systems. For instance, versions of Gottfried Michael Koenig's *Funktion Rot* and Trevor Wishart's *Imago* were produced at the ZKM, in partnership with the composers using the 43.4-channel Klangdom in the Kubus.

More and more institutions are displaying their commitment to surround sound by building concert halls fitted with the requisite HDLAs. The Klangdom at the ZKM; the Lithuanian Academy of Music and Theatre in Vilnius; the Institut für Elektronische Musik und Akustik (IEM) in Graz; the studio of the Center for Computer Research in Music and Acoustics (CCRMA) at Stanford University; the Espace de Projection at the Institut de Recherche et Coordination Acoustique/Musique (IRCAM) in Paris; the AlloSphere at the University of California, Santa Barbara; the auditorium HE 101 at the Technical University of Berlin—and the temporary installations at the Curtis R. Priem Experimental Media and Performing Arts Center (EMPAC) at Rensselaer Polytechnic Institute, the Hochschule Furtwangen University, the Sonic Lab at Queen's University Belfast, the Cube at Virginia Polytechnic Institute and State University, the Société des arts

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technologiques in Montreal, and the Université de Montréal—are only a few examples. Many more facilities are in planning or already under construction. Moreover, surround-system technology has become an integral part of the cinema experience, and has thereby entered mainstream consumer society. One may hope that surround-sound listening technologies for audiophile end users will be developed, as they will facilitate the exchange and the experience of ambitious new surround-sound compositions. This would have the result of making the vast experiential advantages of surround-sound music easier to access. A more lively exchange between artists in the field would also have the advantage of further developing space-based composition and making it more accessible. Many innovative compositions have already served as inspiration for further developments in the use of space. Universal storage methods and cross-platform compatibility should make such works easier to share. A network established with the aim of disseminating spatial music would be a great asset: There is immense potential for development both within the field of experimental music and beyond.

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